

1 Participant Information Sheet

Project title: Software quality analysis of scientific models

Contact XXX if you have any questions regarding this questionnaire and or the study

Context of the Study

Thank you for taking the time to answer this questionnaire and take part in this study. The purpose of this study is to understand the structural robustness of global land use and hydrological models, focusing on how software architecture relates to model stability and maintenance risk. The data collected in this questionnaire will be used to understand how the experience of developers and maintainers of global land use and hydrological models' code compares to our network-based analysis of global land use and hydrological models' call graph. This questionnaire will take an average of 15 to 20 minutes to complete and will close March 18th 2026 at 23:59.

Data Protection and Consent

The project will adhere to all relevant UK data laws. In its collection, storage and dissemination of data, the project will follow the UK General Data Protection Regulation's (GDPR) principles of lawfulness, fairness and transparency (accessible here). The project will adhere to the requirements of the UK Data Protection Act (2018) (accessible here).

Your answers will remain anonymous and your participation is voluntary. As your answers to this questionnaire are anonymous, withdrawal from the questionnaire can only be done by exiting the tab being used before the questionnaire is submitted. Once submitted, data cannot be withdrawn as it will not be identifiable.

The DAWN research group will have access to your anonymous data. If publication occurs of the wider study that includes the results of these questionnaires this anonymous data set will be shared and made publicly available.

Benefits of Participation

After you submit the form, please contact **XXX** citing that you have completed this survey and which model you are associated with if you would like to receive a short summary with the structural metrics we have computed on your model's code and structure, and a short summary comparing your assessments with our results. This summary will be sent once this survey has closed and we have analysed the results. Do not share this data with anyone, the data must not be published or redistributed without our permission. Please do not provide any identifiable information that could connect yourself to your questionnaire answers.

By selecting 'yes' to the below **I consent** question, you:

1. I have read and understood the participation information sheet
2. understand that participation in this questionnaire is completely voluntary.
3. give permission for your answers to be used within this study and academic works
4. understand and agree to the fact that due to the anonymity of this questionnaire, once the questionnaire is submitted, your data cannot be withdrawn
5. understand that your data will be gathered, stored and protected in line with GDPR
6. understand that the short summary that can be received after participation cannot be shared with anyone else, published or redistributed without the permission of DAWN
7. I agree to take part in this study

I consent: **[yes check box]**

2 Eligibility Statement

I confirm that I am involved in the code of a Global Land Use and / or Hydrological models (e.g. as a maintainer, coder, contributor etc.) **[yes check box]**

I am involved in the code of: (if multiple please choose the model you have the most involvement in).

- CTSM
- CWatM
- DBH
- GR4J
- H08
- HydroPy
- HYPE
- MHM
- ORCHIDEE
- PCR-GLOBWB
- SACRAMENTO
- SWAT
- None of the above **If the respondent answers 'None of the above', they are screened out.**

3 Path-level evaluation

For each of the 10 paths:

How familiar are you with the following call sequence?

A → B → C → D

- 0. Not familiar at all.
- 1. Aware of some functions, but not the path.
- 2. Moderately familiar.
- 3. Very familiar.

If the respondent answers 'Not familiar' we skip evaluations for this path and move to the next.

3.1 Evaluation questions

1. The path corresponds to a real execution chain in a typical model run
2. The path traverses routines that are central to the model's main workflow
3. The path is visited often by developers for bug fixes, refactoring or adjustments
4. If a fault were to occur in the path, it would plausibly have system-wide consequences (e.g., crashes, widespread anomalies)

Possible answers:

- 1. Strongly disagree
- 2. Disagree
- 3. Neither agree nor disagree
- 4. Agree
- 5. Strongly Agree

5. Do you have any additional information about this path? (Please do not include any personal information)